

6 Simplify these expressions.
Leave your answer in index form.

A $3^4 \div 3^2 =$ **B** $2^3 \div 2^2 =$

C $4^6 \div 4^3 =$ **D** $3^5 \div 3^2 =$

E $2^7 \div 2^3 =$ **F** $5^5 \div 5^3 =$

G $3^4 \div 3^1 =$ **H** $2^6 \div 2^2 =$

I $4^5 \div 4^3 =$ **J** $5^4 \div 5^3 =$

K $2^9 \div 2^5 =$ **L** $3^8 \div 3^2 =$

M $5^7 \div 5^3 =$ **N** $7^5 \div 7^2 =$

O $8^7 \div 8^2 =$

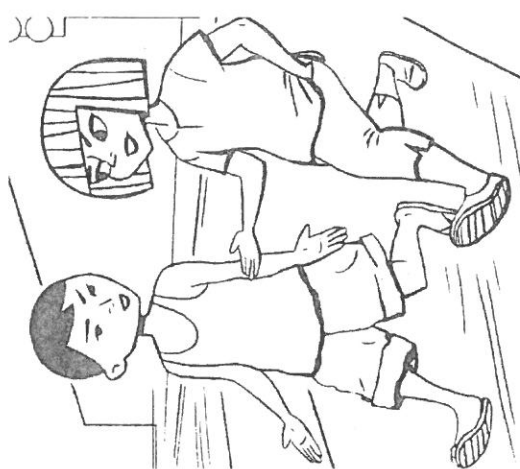
P $6^5 \div 6^3 =$

Q $7^6 \div 7^3 =$

R $10^7 \div 10^5 =$

S $9^6 \div 9^5 =$

T $8^{12} \div 8^9 =$



3 Simplify these expressions.
Leave your answer in index form.

A $3^2 \times 3^4 =$ **B** $2^3 \times 2^2 =$

C $4^2 \times 4^3 =$ **D** $3^5 \times 3^2 =$

E $2^3 \times 2^3 =$ **F** $5^3 \times 5^2 =$

G $3^4 \times 3^5 =$ **H** $2^6 \times 2^2 =$

I $4^3 \times 4^5 =$ **J** $5^4 \times 5^3 =$

K $2^5 \times 2^7 =$ **L** $3^4 \times 3^1 =$

M $5^4 \times 5^6 =$

N $7^3 \times 7^2 =$

O $4^4 \times 4^3 =$

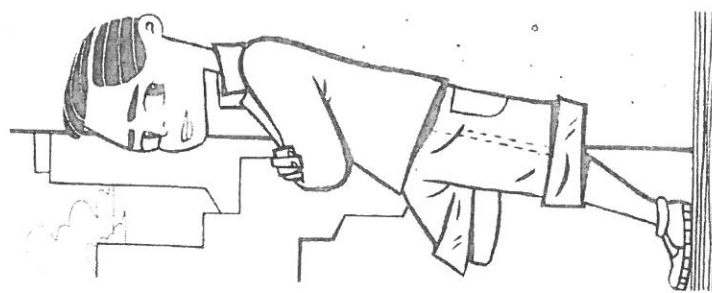
P $6^5 \times 6^4 =$

Q $7^2 \times 7^5 =$

R $10^3 \times 10^5 =$

S $2^5 \times 2^3 \times 2^4 =$

T $5^2 \times 5^3 \times 5^2 =$



Investigating the index rules 2A

A power of a power

- 1** Simplify $(52^2)^3$. Leave your answer in index form.
Hint: $(52^2)^3$ means three lots of 52^2 are multiplied together.

$$(52^2)^3 = 52 \times 52 \times 52$$

$$=$$

$$=$$

$$=$$

- 2** Simplify $(2^4)^2$. Leave your answer in index form.

$$(2^4)^2 = 2^4 \times 2^4$$

$$=$$

$$=$$

$$=$$

Can you see a short way to do these questions?

Anthony suggested:



*You multiply the indices.
 The base number does not change.*

Anthony's rule is correct.

Rule: To find a power of a power:

Multiply the indices.
 The base number does not change.

INDICES

UNIT 1 TOPIC 2

- 3** Simplify these expressions.

Leave your answer in index form.

A $(3^4)^2 =$ **B** $(2^3)^2 =$

C $(4^2)^3 =$ **D** $(3^5)^2 =$

E $(2^3)^4 =$ **F** $(5^2)^4 =$

G $(3^4)^5 =$ **H** $(2^6)^2 =$

I $(4^5)^3 =$ **J** $(5^4)^3 =$

K $(2^4)^5 =$ **L** $(3^8)^2 =$

M $(5^7)^3 =$ **N** $(7^5)^4 =$

O $(8^7)^2 =$ **P** $(6^4)^5 =$

Q $(7^6)^3 =$ **R** $(6^7)^5 =$

S $(9^6)^5 =$ **T** $(2^5)^9 =$

Now use your calculator to check your answers for parts A to F.

